Assignment: Image Generation by Stable Diffusion

Task 1: Text-to-Image Generation

Develop a Stable Diffusion–based text-to-image generation system.

Requirements

- Use a pretrained Stable Diffusion model as the starting point.
- Prepare a dataset containing image–text pairs.
- Fine-tune the model using the prepared dataset.
- Use textual prompts to generate images.
- Generate at least **10 images using different prompts**.
- Compare the generated images with the corresponding target images.

Input

- **x:** Text prompt
- **y:** Target image

Example:

> x = "A red sports car parked beside a lake at sunset"
> y = corresponding target image

Evaluation

Evaluate the generated images using:

- CLIP similarity
- FID
- SSIM, where an appropriate reference image exists
- Visual inspection

Discuss how prompt complexity affects image quality and semantic consistency.

Task 2: Image-to-Image Generation

Develop an image-to-image generation system using Stable Diffusion.

Requirements

- Use an input image together with a text prompt.
- Generate modified versions of the input image.
- Experiment with different **denoising strength** values.
- Use at least **10 input images**.
- Generate multiple outputs for selected input images.

Input

- **x:** Input image + text prompt
- **y:** Target/desired image

Example:

> x = photograph of a house + "A house in a snowy mountain landscape"
> y = corresponding desired image

Experiments

Investigate the effect of:

- Denoising strength
- Guidance scale
- Number of inference steps
- Different text prompts

Evaluation

Evaluate:

- Semantic similarity with the prompt
- Similarity to the input image
- CLIP score
- FID, where a suitable reference dataset is available
- Visual quality

Discuss the trade-off between **preserving the input image** and **following the text prompt**.

Task 3: Image Inpainting

Implement image inpainting using Stable Diffusion.

Requirements

- Prepare images containing objects or regions to be removed/replaced.
- Create a binary mask indicating the region to be modified.

- Provide a text prompt describing the desired content.
- Generate the completed image.
- Perform experiments using at least **10 images**.

Input

- **x:** Original image + mask + text prompt
- **y:** Target completed image

Example:

> Original image: a person standing in a garden
> Mask: region containing the person
> Prompt: "A beautiful garden with flowers and trees"

The system should replace the masked region while preserving the unmasked region.

Evaluation

Evaluate:

- Quality of the generated region
- Consistency between generated and surrounding regions
- CLIP similarity
- SSIM/PSNR against ground truth, when available
- Human/visual evaluation

Compare the results obtained using different mask sizes and shapes.

Task 4: Image Outpainting

Develop an outpainting system that extends an image beyond its original boundaries.

Requirements

- Select at least **10 images**.
- Extend the image in one or more directions.
- Create a larger canvas containing the original image and an empty region.
- Use an appropriate mask to identify the region to be generated.
- Generate the missing region using Stable Diffusion.
- Experiment with different prompts and generation parameters.

Input

- **x:** Original image + outpainting mask + text prompt

- **y:** Extended image

Example:

> Original image: a small landscape photograph
> Mask: newly added region around the original image
> Prompt: "A wide mountain landscape with blue sky and clouds"

Common Experimental Requirements

For all four tasks:

1. Model

Use a publicly available pretrained Stable Diffusion model. Clearly report:

- Model name/version
- VAE
- Text encoder
- UNet/diffusion component
- Image resolution
- Number of parameters, if available

2. Dataset

Use the **dataset prepared collectively by all of you** for this assignment. Each student should use the common dataset prepared by the class rather than selecting a separate dataset.

Describe:

- Number of images
- Image resolution
- Text annotations/captions
- Training/validation/test split
- Preprocessing and augmentation
- How the dataset was prepared and divided among the four tasks

3. Training/Fine-Tuning

Document:

- Learning rate

- Batch size
- Number of epochs/steps
- Optimizer
- Scheduler
- Noise schedule
- Image resolution
- Hardware/GPU
- Training time

Explain which components are trained and which components are frozen.

4. Evaluation

Use both quantitative and qualitative evaluation.

Required Experiments

Students must investigate the **effect of following parameters**, such as:

- Number of inference steps
- Denoising strength
- Image resolution
- Random seed
- Prompt formulation

For each parameter, keep the other parameters fixed and compare the generated results.

Important Question

For each of the four tasks, clearly identify:

What is the input condition x, and what is the target y during training/fine-tuning?

Students should explain why the conditioning information differs among:

- Text-to-image
- Image-to-image
- Inpainting
- Outpainting

The assignment should emphasize that these four tasks are **not trained with exactly the same input-conditioning structure**, even though they can use the same underlying diffusion framework.

Final Report

The report must contain:

1. Introduction
2. Stable Diffusion Architecture
3. Dataset Description
4. Methodology
5. Training/Fine-Tuning Procedure
6. Experimental Setup
7. Results
8. Quantitative Evaluation
9. Qualitative Evaluation
10. Parameter Analysis
11. Comparison of Text-to-Image, Image-to_Image, Inpainting and Outpainting
12. Discussion
13. Limitations
14. Conclusion
15. References

Include generated images, input images, masks, prompts, training curves, evaluation scores, and comparison tables.